06. Model Selection

Explain content-based filtering, bag-of-words and cosine similarity.

Project: Movie Recommender System

Source archive supplied in this conversation: Movie-recommender-system-main.zip | Notebook: notebook.ipynb | App: app.py

Content-based filtering

A content-based recommender suggests items similar to an item or profile based on item attributes. It does not need other users' ratings. This project uses a movie's metadata as its content representation.

CountVectorizer

Let the vocabulary contain V terms. Each movie is represented as $x \in R^V$, where x_j is the count of vocabulary term j in that movie's tags. The notebook caps V at 5,000 and removes English stop words. This is a bag-of-words representation: word order is ignored.

Cosine similarity

For vectors x and y, cosine similarity is $(x \cdot y)/(||x||\ ||y||)$. It measures the angle between vectors and ranges from –1 to 1 in general; for non-negative count vectors it lies between 0 and 1. A score of 1 means the vectors point in the same direction; 0 means no shared positive dimensions.

Why cosine rather than Euclidean distance

Cosine focuses on relative term composition and is less sensitive to document length than raw Euclidean distance. Two movies with similar tag proportions can be close even if one has a longer overview and therefore larger raw counts.

Alternatives

- TF-IDF + cosine: reduce the influence of common terms.
- BM25: retrieval-oriented term weighting, often useful for text search.
- Word/document embeddings: capture semantic similarity beyond exact token overlap.
- Hybrid recommendation: blend content similarity with collaborative filtering, popularity, recency and user context.
- Learning-to-rank: train a ranking model from user interactions and relevance labels.

Why not collaborative filtering?

The repository does not contain user IDs, ratings or user-item interactions. Collaborative filtering cannot be trained from the included movie metadata alone. A hybrid system could be added if interaction data became available.